Exoskeleton Control System

System Design Document

AutomationForExoskeleton

Version 1.3 — April 5, 2026

1 Executive Summary

This document describes the architecture of the software-centric control module for the ExoTechHK lower-limb exoskeleton. The system integrates inertial measurement units (IMUs), Support Vector Machine (SVM) classification, and optional Kalman Filter visualization to provide real-time, adaptive assistance.

Version 1.3 uses **HuGaDB** from **GitHub v1** only, but now treats it as a protocol-aware and corruption-aware dataset: `LoadHuGaDB.m` reads raw `.txt` files, applies the official gyro corruption matrix (CSV next to the loader), preserves `single_activity` versus `multi_activity_sequence` metadata, and stores quality flags in the cache. Binary and multiclass HuGaDB training/evaluation default to excluding `ExoConfig.HUGADB.HELDOUT_SUBJECTS`, filtering to `ExoConfig.HUGADB.DEFAULT_PROTOCOLS`, and skipping invalid sessions/windows so the policy matches replay simulation. The common feature vector remains **30-dimensional** (six IMU slots × five statistics per slot), with **binary** and **multiclass** ECOC SVM paths and the asymmetric finite state machine for assist gating.

2 System Architecture

2.1 Configuration (`ExoConfig.m`)

The system is driven by a configuration class that centralizes tuning parameters.

- **Sampling Rate:** 100 Hz
- **Window Size:** 1.0 s (100 samples)
- **Step Size:** 0.5 s (50 samples)
- **Paths:** `usc_had_dataset.mat`, `hugadb_dataset.mat`, `Binary_SVM_Model_hugadb_only.mat`, `Multiclass_SVM_ECOC_usc_had.mat`, `Multiclass_SVM_ECOC_hugadb.mat`
- **LOCOMOTION:** `N_IMU_SLOTS = 6`, `FEATURES_PER_IMU = 5` (must match `Features.m`)

2.2 Data Flow Pipeline

1. **Input:** Public `.mat` caches via loaders; held-out HuGaDB sessions replay through `LoadHuGaDBSimulationData.m`.
2. **Pre-processing:** HuGaDB cache is normalized to m/s^2 and rad/s during loading, tagged by session protocol, and screened by a HuGaDB quality gate before windows are used for training/evaluation.

3. **Feature extraction:** Five features per IMU (`Features.m`). HuGaDB uses **six IMUs** per window (`FeaturesFromImuStack`). USC-HAD uses **one IMU + zero padding** (`LocomotionFeatureVector`) for benchmark comparisons.
4. **Inference — binary:** RBF SVM (`fitcsvm`) → class 0/1 (inactive vs active movement per dataset label maps).
5. **Inference — multiclass (optional):** ECOC SVM (`fitcecoc`) on **native** labels: USC-HAD **12** trial activities or HuGaDB **12** per-sample IDs (`ActivityClassRegistry.m`). Exo assist maps active-movement classes through `RealtimeFsmFromActivityClass.m` (dataset argument: `'usc_had'` / `'hugadb'`).
6. **State machine:** `RealtimeFsm.m` applies asymmetric hysteresis on **binary** predictions.
7. **Fusion:** Optional Kalman (`imufilter`) computes a segment-pitch visualization trace in parallel.
8. **Output:** Control command (0/1) + visualization trace; multiclass scripts add activity labels for logging/plots.

3 Core Components

3.1 Feature extraction (`Features.m`, `FeaturesFromImuStack.m`, `LocomotionFeatureVector.m`)

Per IMU window, the feature vector is **1×5**:

1. Mean acc magnitude
2. Variance of acc magnitude
3. Mean gyro magnitude
4. Variance of gyro magnitude
5. Dominant FFT frequency of vertical acceleration

Training on public data: HuGaDB trials store **N×3×6** acc/gyro (`LoadHuGaDB.m`); stacked features are **1×30**. USC-HAD trials use one IMU → **1×5** inside `LocomotionFeatureVector` plus **25 zeros**.

3.2 Activity labels (`ActivityClassRegistry.m`)

Multiclass ECOC uses **dataset-native** IDs (no cross-dataset unification): USC-HAD **1...12** per trial, HuGaDB **1...12** per sample (window label = mode over samples). Binary training uses dataset-specific active / non-active ID groups from `ExoConfig.DS`.

3.3 Sensor fusion (`FusionKalman.m`)

Static class wrapping MATLAB `imufilter`. Under the HuGaDB replay path it produces an orientation-derived segment-pitch trace for visualization.

3.4 Finite State Machine (`RealtimeFsm.m`)

Asymmetric hysteresis: **3** consecutive active binary predictions to enter assist; **5** consecutive inactive predictions to exit.

4 Execution & File Organization

4.1 Path management

Main scripts resolve `projectRoot` locally and add `config/`, the active `scripts/` folder, and `src/` (recursive) as needed. Dataset loader functions remain under `data/USC-HAD/`, `data/HuGaDB/`, etc., and can be added to the MATLAB path when rebuilding local caches.

USC-HAD: `LoadUSCHAD.m` → `data/USC-HAD/usc_had_dataset.mat` (recursive raw `.mat` under `USC-HAD_raw/`).

HuGaDB: `LoadHuGaDB.m` → `data/HuGaDB/hugadb_dataset.mat` (GitHub v1 under `v1_cleanup_github/` flat layout or legacy nested `v1_raw`/flat paths; official gyro corruption matrix zeroes bad IMU gyros; `readmatrix` treats `na` as missing when present; all-six-corrupt sessions are retained as accelerometer-only samples so classes such as bicycling / sitting-in-car are not lost; six IMUs per row; optional manifest adds `huGaDBProvenance` / **`huGaDBSessionProtocol`**; accepted sessions carry quality metadata and downstream prep functions skip invalid sessions/windows with logged reasons).

Binary training / evaluation: `TrainSvmBinary.m`, `EvaluateSvmConfusion.m`, `RunSvmDatasetAblation.m` (USC-HAD and HuGaDB).

Multiclass: `PrepareTrainingDataMulticlass.m`, `TrainSvmMulticlass.m`, `EvaluateMulticlassConfusion.m`, `RunTrainEvalMulticlass.m`, `RunExoskeletonPipelineMulticlass.m`.

CUHK-REXYIM (local captures): `ImportData.m` → `data/CUHK-REXYIM/<activity>/` with `Accelerometer.csv`, `Gyroscope.csv`, optional `Annotation.csv`.

4.2 Directory structure (abridged)

```
AutomationForExoskeleton/
  config/                 # ExoConfig.m, ActivityClassRegistry.m
  data/USC-HAD/           # LoadUSCHAD.m, usc_had_dataset.mat
  data/HuGaDB/            # LoadHuGaDB.m, hugadb_dataset.mat
  data/CUHK-REXYIM/       # project phone CSVs per activity (ImportData.m)
  models/                 # Binary_SVM_Model*.mat, Multiclass_SVM_ECOC_*.mat
  results/                # figures/ and metrics/ outputs
  scripts/                # Train/eval/pipeline scripts
  src/
    acquisition/          # PrepareTrainingData*.m
    classification/       # RealtimeFsm.m, RealtimeFsmFromActivityClass.m
    features/             # Features.m, FeaturesFromImuStack.m, ...
    fusion/
```